State-Dependent Uncertainty Bounds for Robust Output-Feedback MPC

Derivation and a Falcon Certificate-Reuse Numerical Example

Research draft

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I Introduction

Robust output-feedback MPC must account for both uncertain state estimates and deviations between an observer trajectory and a nominal prediction. A global uncertainty envelope can be unnecessarily conservative in operating regions with smaller model mismatch. Our objective is to retain an existing offline design while allowing the predicted uncertainty allowance to depend on the operating region covered by the error tubes.

The organization follows [1]: preliminaries, estimation bounds, robust output-feedback MPC, and a numerical example. Joint nominal and tube prediction, a free nominal initial center, and shifted-candidate arguments are established ingredients of Section IV-B of that reference. They are not claimed as new here. Our proposed extension concerns the local process-envelope construction and its incorporation into coupled continuous-time radius dynamics.

We consider robust output-feedback MPC with fixed metrics and observer/controller gains, a fixed regulation target, and time-invariant constraints.

II Preliminaries

A Plant, observer, and nominal prediction

Consider

$$\dot{x} = f_0(x,u) + E_0 w_x, \qquad w_x \in \mathcal{W}(x,u), \tag{II.1a}$$

$$y = Cx + F\eta, \qquad \eta \in \mathcal{N}, \tag{II.1b}$$

$$\dot{\hat{x}} = f_0(\hat{x},u) + L(y - C\hat{x}), \tag{II.1c}$$

$$\dot{z} = f_0(z,v), \qquad u = v + K(\hat{x} - z). \tag{II.1d}$$

Here $x, \hat{x}, z \in \mathbb{R}^n$ and $u, v \in \mathbb{R}^{n_u}$; v is the nominal control input, $E_0 \in \mathbb{R}^{n\times m}$ maps process-disturbance components into the state dynamics, and $\mathcal{W}(x,u) \subset \mathbb{R}^m$ describes their admissible values at the current state and input. The physical process disturbance is $d = E_0 w_x$. Measurement noise belongs to the fixed bounded set $\mathcal{N}$, and L is a fixed observer gain. The nominal model f_0 includes any identified constant bias, so w_x represents the centered residual uncertainty.

We assume a global baseline box encloses the process uncertainty on the operating region under consideration:

$$\mathcal{W}(x,u) \subseteq \mathcal{W}_0, \qquad \mathcal{W}_0 = \{w \in \mathbb{R}^m : |w_j| \le \bar{w}_j,\ j = 1,\dots,m\}, \quad \bar{w}_j > 0.$$

The local sets $\mathcal{W}(x,u)$ specify the state–input dependence. Their containment in $\mathcal{W}_0$ permits use of a global baseline certificate. Section III-B introduces the particular family of local sets used to obtain an explicit certificate reduction. Let $P = P_\delta \succ 0$, $X = P^{-1}$, and $T^\top T = P$. Define

$$e = x - \hat{x}, \quad \delta = \hat{x} - z, \quad V_e = e^\top P e, \quad V_e \le R, \quad \|\delta\|_P \le s. \tag{II.2}$$

R is a squared radius and $r = \sqrt{R}$ is a radius; neither denotes the true estimation error. The symbol P is distinct from a terminal-cost matrix. Since $x - z = e + \delta$,

$$\|x - z\|_P \le \sqrt{R} + s. \tag{II.3}$$

Subtracting the dynamics gives

$$\dot{e} = f_0(x, u) - f_0(\hat{x}, u) - LCe + E_0 w_x - LF\eta, \tag{II.4a}$$
$$\dot{\delta} = f_0(\hat{x}, u) - f_0(z, v) + LCe + LF\eta. \tag{II.4b}$$

Process disturbance enters the estimation error directly. It influences observer-to-nominal tracking indirectly through e; the observer equation itself contains no additive plant process disturbance.

B Time and prediction notation

Actual signals use a single time subscript, for example $x_t, \hat{x}_t, u_t$, with continuous physical time t. Plans are computed at sampling instants $t \in \{0, h, 2h, \ldots\}$, where $h > 0$ is the sampling period. Predicted trajectories use paired subscripts: $z_{\tau|t}, v_{\tau|t}, R_{\tau|t}, s_{\tau|t}$ denote predictions computed at time t for physical time $t + \tau$, with continuous prediction time $\tau \in [0, H_p]$ and horizon $H_p \ge h$. A dot denotes differentiation with respect to physical time for actual signals and with respect to τ, holding t fixed, for predictions. The next update occurs at $t + h$. The symbol $\bar{R}_0$ denotes the externally supplied initial squared estimation-error bound. Time subscripts are omitted in pointwise derivations when no time dependence needs to be displayed. Parentheses denote function evaluation, such as $f_0(x_t, u_t)$.

C Well-posedness

Assumption 1 (Well-posed model and observer)**.** *The nominal model f_0 is continuously differentiable in state and input. Given initial states and admissible locally bounded measurable disturbance and control signals, the plant, observer, and nominal dynamics admit unique absolutely continuous solutions over the intervals considered.*

This assumption ensures that the state trajectories used in the error analysis are uniquely defined for the specified initial conditions and admissible signals.

III State-Dependent Estimation Error Bounds

A Recovering the global quadratic certificate

LMIs and error inequality. The starting point is the offline robust output-feedback design of Benders et al. [2]. Let $\epsilon \ge 0$ denote the constant estimation-error radius supplied by its global observer certificate, expressed here in the P-norm:

$$\|x - \hat{x}\|_P \le \epsilon, \qquad \text{equivalently} \qquad V_e \le \epsilon^2.$$

This invariant bound holds provided $V_{e,0} \le \epsilon^2$ and the certificate assumptions hold. This subsection recovers the original constant radius ϵ and introduces a time-varying squared radius R_t; Section III-B then constructs the local process-envelope reduction. For $b \in \mathbb{R}^n$, introduce the symmetric off-diagonal block

$$\mathcal{S}(b) = \begin{bmatrix} 0 & b \\ b^\top & 0 \end{bmatrix}. \tag{III.1}$$

Let $A_o = A - LC$ and $q_0 = \lambda\epsilon^2$, with $\lambda > 0$. Let W and H be symmetric process and noise envelope matrices, respectively. The following observer-certificate LMIs are taken from the

ROHMPC implementation accompanying Benders et al. [2, 3], specialized to a constant metric and expressed in the present notation. They are sufficient design conditions for $\dot{V}_e + \lambda V_e \le q_0$:

$$\mathcal{S}(E_0 w) \preceq W \qquad \forall w \in \mathcal{W}_0, \tag{III.2a}$$

$$\mathcal{S}(-LF\eta) \preceq H \qquad \forall \eta \in \mathcal{N}, \tag{III.2b}$$

$$W + H + \begin{bmatrix} XA_o^\top + A_o X + \lambda X & 0 \\ 0 & -q_0 \end{bmatrix} \preceq 0. \tag{III.2c}$$

Here $A = \partial f_0/\partial x$. The last inequality is a condition on the state Jacobians, and must hold along the segments joining the true and estimated states at their common input. Since X is constant, its time derivative is zero.

To see explicitly how these LMIs imply the error bound, first use the global baseline uncertainty $w_x \in \mathcal{W}_0$, writing $w = w_x$. Equation (II.4) can be written as

$$\dot{e} = A_o e + b, \qquad b := E_0 w - LF\eta.$$

For the nonlinear model, the matrix in the error dynamics is the segment average

$$A_o = \int_0^1 \frac{\partial f_0}{\partial x}(\hat{x} + \theta e, u)\, d\theta - LC.$$

Indeed, integrating the state Jacobian along the segment gives $f_0(x,u) - f_0(\hat{x},u) = (A_o + LC)e$. Since the dynamics LMI is affine in the Jacobian, its validity along the segment implies its validity for this average. The vector b collects the process disturbance and measurement-noise contributions.

Adding the two envelope inequalities and the dynamics LMI cancels W, H. Evaluating the resulting quadratic form at $[Pe; 1]$ gives

$$\dot{V}_e \le -\lambda V_e + q_0. \tag{III.3}$$

The expanded calculation and equivalent congruence transformation are given in Appendix A.

Comparison equation and radius propagation. To bound the estimation error, introduce a scalar squared radius R_t defining the time-varying error set

$$\mathcal{E}_t = \{e : e^\top P e \le R_t\}.$$

Thus R_t is the upper boundary in the scalar V_e coordinate, and $e^\top P e = R_t$ is the corresponding boundary in error-state space. Choose $R_0 \ge V_{e,0}$ and propagate this boundary using the comparison equation, whose solution bounds V_e:

$$\dot{R} = -\lambda R + q_0. \tag{III.4}$$

Subtracting the comparison equation (III.4) from (III.3) gives

$$\frac{d}{dt}(V_e - R) \le -\lambda(V_e - R), \qquad V_{e,t} - R_t \le e^{-\lambda t}(V_{e,0} - R_0) \le 0.$$

Consequently, $V_{e,t} \le R_t$ and $\sqrt{R_t}$ bounds the estimation error in the P-norm wherever the derivative inequality holds. The comparison equation (III.4) has solution

$$R_t = \epsilon^2 + (R_0 - \epsilon^2)e^{-\lambda t}. \tag{III.5}$$

When $R_0 = \epsilon^2$, it exactly recovers the original constant radius ϵ.

B Local process-envelope reduction

The objective is to replace the global observer forcing q_0 by a smaller allowance that remains valid for the local process set. Starting from the model $d = E_0 w_x$, $w_x \in \mathcal{W}(x,u)$, we now specialize to a family parameterized by a scalar amplitude:

$$\mathcal{W}(x,u) = \mathcal{W}(x) = D(a(x))\mathcal{W}_0, \qquad \beta \le a(x) \le 1, \quad 0 < \beta < 1. \tag{III.6}$$

Thus $w_x = D(a(x))w$ for some $w \in \mathcal{W}_0$. Choose a fixed subset $\mathcal{J} \subseteq \{1,\dots,m\}$ identifying the disturbance components assumed to vary with the operating state; the remaining components retain their global bounds. This is a modeling choice, and selecting all components is permitted. Define

$$D(a) = \operatorname{diag}(d_1(a),\dots,d_m(a)), \qquad d_j(a) = \begin{cases} a, & j \in \mathcal{J}, \\ 1, & j \notin \mathcal{J}. \end{cases}$$

This construction is affine in a, has $D(1) = I$, and produces nested sets:

$$\beta \le a_1 \le a_2 \le 1 \quad \Longrightarrow \quad E_0 D(a_1)\mathcal{W}_0 \subseteq E_0 D(a_2)\mathcal{W}_0. \tag{III.7}$$

The inclusion follows from the centered box bounds $|w_j| \le \bar{w}_j$: increasing a enlarges the bounds of the selected components while leaving the others fixed. It remains valid after applying E_0.

The scalar parameter serves two purposes: affine $D(a)$ makes the vertex quadratics defined before (III.10) convex in a, enabling the endpoint chord bound; nested disturbance sets allow an upper bound on $a(x)$ to cover the predicted tube.

A general matrix $D(x)$ can also describe state-dependent uncertainty, but need not have these properties. Its disturbance envelope would require a different bounding argument. Section III-C specifies the velocity-dependent choice of $a(x)$.

Let $W \in \mathbb{S}^{n+1}$ satisfy the baseline process-envelope inequality (III.2a), where $\mathbb{S}^{n+1}$ denotes the symmetric $(n+1)\times(n+1)$ matrices. Introduce block notation for this envelope matrix:

$$W = \begin{bmatrix} Q & h_w \\ h_w^\top & c \end{bmatrix}, \qquad Q \succ 0. \tag{III.8}$$

Here $Q \in \mathbb{S}^n$ is the upper-left block, $h_w \in \mathbb{R}^n$ is the off-diagonal block, and $c \in \mathbb{R}$ is the bottom-right entry. These are fixed blocks of the global baseline envelope matrix W; the subscript in h_w does not indicate dependence on the disturbance w. The local construction assumes $Q \succ 0$, keeps Q and h_w fixed, and reduces the scalar entry c to accommodate smaller disturbance sets.

Taking the Schur complement with respect to $Q \succ 0$ gives

$$\mathcal{S}(E_0 D(a) w) \preceq W \iff (E_0 D(a) w - h_w)^\top Q^{-1} (E_0 D(a) w - h_w) \le c. \tag{III.9}$$

The matrix Q^{-1} determines the quadratic weighting, h_w its center, and c the admissible level. The box $\mathcal{W}_0$ has 2^m vertices $w^{(i)}$, obtained by choosing each component as $\pm\bar{w}_j$. Define

$$\phi(w,a) := (E_0 D(a) w - h_w)^\top Q^{-1} (E_0 D(a) w - h_w), \qquad \phi_i(a) := \phi(w^{(i)}, a).$$

Here ϕ_i is the expression at the fixed vertex $w^{(i)}$, regarded as a function of the scale a. Their pointwise maximum is

$$g(a) = \max_{i=1,\dots,2^m} \phi_i(a) = \max_{w \in \operatorname{vert}(\mathcal{W}_0)} \phi(w,a). \tag{III.10}$$

For fixed a, $\phi(\cdot, a)$ is convex, so its maximum over the box is attained at a vertex: $g(a) = \max_{w\in\mathcal{W}_0} \phi(w,a)$. Affine dependence of $D(a)$ makes each $\phi_i(a)$ convex in a; hence their maximum g is convex. Nestedness also gives $g(\beta) \le g(1)$. A convex function lies below its endpoint chord. Consequently,

$$g(a) \le g(1) - \frac{1-a}{1-\beta}(g(1) - g(\beta)). \tag{III.11}$$

The convexity calculations, geometric interpretation, and secant-slope comparison are given in Appendix B. The following proposition transfers this reduction to the process-envelope entry c and the observer forcing q_0 by reducing both by the same amount.

Proposition 1 (Local forcing from a fixed certificate)**.** *Suppose the baseline certificate* (III.2) *holds exactly,* $Q \succ 0$*, and the local disturbance sets have the structure* (III.6)*. Choose* $0 \le \Delta \le g(1) - g(\beta)$ *and define*

$$\theta(a) = \frac{1-a}{1-\beta}, \tag{III.12}$$

$$W(a) = \begin{bmatrix} Q & h_w \\ h_w^\top & c - \Delta\theta(a) \end{bmatrix}, \tag{III.13}$$

$$q(a) = q_0 - \Delta\theta(a). \tag{III.14}$$

For every $a \in [\beta, 1]$*, the reduced envelope satisfies* $\mathcal{S}(E_0 D(a) w) \preceq W(a)$ *for all* $w \in \mathcal{W}_0$*. Together with the unchanged noise envelope, this gives the derivative bound* $\dot{V}_e \le -\lambda V_e + q(a)$*.*

Proof.

Step 1: Validity of the reduced process envelope.

At $a = 1$, $D(1) = I$. The process-envelope inequality (III.2a) of the global baseline certificate, together with the Schur-complement equivalence (III.9) and the vertex-maximum identity (III.10), implies

$$c \ge \max_{w \in \mathcal{W}_0} \phi(w, 1) = \max_i \phi_i(1) = g(1).$$

Equation (III.11) and the choice of Δ give $g(a) \le c - \Delta\theta(a)$, proving the local envelope inequality by (III.9).

Step 2: Preservation of the observer certificate.

To retain the dynamics certificate, reduce q_0 by the same amount as the bottom-right entry of W. Indeed, writing $B_o = X A_o^\top + A_o X + \lambda X$ gives

$$W(a) + H + \operatorname{diag}(B_o, -q(a)) = W + H + \operatorname{diag}(B_o, -q_0) \preceq 0.$$

Thus the disturbance envelope can shrink while the dynamics LMI remains identical. Adding the reduced process-envelope inequality and the unchanged noise-envelope inequality (III.2b), together with this dynamics inequality, yields $\dot{V}_e \le -\lambda V_e + q(a)$ by the quadratic-form calculation in Appendix A. □

At $a = 1$, $W(a) = W$ and $q(a) = q_0$. For smaller a, the chord bound gives a guaranteed process-envelope reduction, which is transferred to the observer forcing without changing the metric or gains. The construction therefore follows the sequence

$$\mathcal{W}(x) = D(a(x))\mathcal{W}_0 \;\longrightarrow\; g(a) \;\longrightarrow\; W(a) \;\longrightarrow\; q(a).$$

Here $\mathcal{W}(x)$ is a set of admissible disturbance components, $E_0 w_x$ is the physical disturbance, $W(a)$ is its auxiliary matrix envelope, and $q(a)$ is the scalar forcing allowance in the estimation-error inequality. For use in MPC, x is unknown. We must next bound $a(x)$ over the predicted error tube so that the same local forcing is valid for every true state represented by that tube. For the positive-forcing formulation below, choose Δ such that $q_0 - \Delta > 0$.

C Velocity-Dependent Disturbance Bound over the Predicted Tube

Let S_v select the translational velocity components of the state, and denote the true velocity vector by $v_x := S_v x$. The symbol v in the nominal dynamics continues to denote the nominal control input. Consider the velocity-dependent amplitude law

$$a(x) = \beta + (1-\beta)\frac{\|S_v x\|_2^2}{v_{\max}^2}, \qquad \|S_v x\|_2 \le v_{\max}. \tag{III.15}$$

The amplitude enters the physical process disturbance through

$$d(x,w) = E_0 D(a(x))w, \qquad w \in \mathcal{W}_0.$$

The matrix $D(a)$ is the component scaling defined in Section B. Thus a bound on true speed determines an enclosing local process-disturbance set.

Recall that $T^\top T = P$, so $\|e\|_P = \|Te\|_2$. Define

$$c_v := \left\| S_v T^{-1} \right\|_2. \tag{III.16}$$

The induced norm gives $\|S_v e\|_2 \le c_v\sqrt{R}$ and $\|S_v\delta\|_2 \le c_v s$. Since $x = z + \delta + e$, the triangle inequality yields

$$\|S_v x\|_2 \le \|S_v z\|_2 + \|S_v\delta\|_2 + \|S_v e\|_2 \le \|S_v z\|_2 + c_v(s + \sqrt{R}).$$

The intermediate norm calculations are given in Appendix C. Combining this tube bound with the certified domain bound $\|S_v x\|_2 \le v_{\max}$ defines

$$v_{\mathrm{up}} := \min\{v_{\max}, \|S_v z\|_2 + c_v(\sqrt{R} + s)\}. \tag{III.17}$$

Thus v_{up} bounds true speed for every state in the predicted error tube that lies in the certified domain. The cap $v_{\max}$ requires domain containment of the true-state tube; a bound on the nominal speed alone does not establish it.

Since the amplitude law (III.15) is nondecreasing in speed,

$$a(x) \le a_{\mathrm{up}} := \beta + (1-\beta)(v_{\mathrm{up}}/v_{\max})^2. \tag{III.18}$$

The set inclusion (III.7) then gives

$$E_0 D(a(x))\mathcal{W}_0 \subseteq E_0 D(a_{\mathrm{up}})\mathcal{W}_0.$$

Applying Proposition 1 to this enclosing set yields

$$\dot{V}_e \le -\lambda V_e + q(a_{\mathrm{up}}), \qquad q(z,R,s) := q(a_{\mathrm{up}}). \tag{III.19}$$

More explicitly, the scalar used in the estimation-radius dynamics is

$$q(z,R,s) = q_0 - \Delta\frac{1 - a_{\mathrm{up}}(z,R,s)}{1-\beta}.$$

The local uncertainty model specifies $a(x)$, while the predicted state and radii determine its enclosing value a_{up}. The resulting $q(z,R,s)$ is therefore a bound valid throughout that tube. A separate-norm construction can be more conservative because it sums individually maximized contributions and need not recover the original SDP radius at $a = 1$. Appendix D gives this comparison using the same model, metric, and gains.

IV Robust Output-Feedback MPC

A Coupled tube dynamics

Suppose the tracking contraction condition holds along every state–input segment joining (z, v) to $(\hat{x}, v + K(\hat{x} - z))$ in the certified domain:

$$(A + BK)^\top P + P(A + BK) \preceq -2\rho P, \qquad \rho > 0. \tag{IV.1}$$

With $T^\top T = P$, define

$$\gamma := \left\| TLCT^{-1} \right\|_2, \qquad b_\eta := \max_{\eta \in \mathcal{N}} \|TLF\eta\|_2 .$$

The constant b_η bounds the measurement-noise contribution $LF\eta$ to the observer dynamics in the P-norm. Equation (II.4) then implies

$$\frac{d}{dt} \|\delta\|_P \le -\rho \|\delta\|_P + \gamma\sqrt{V_e} + b_\eta. \tag{IV.2}$$

The derivation of (IV.2), using the contraction condition (IV.1) and the Cauchy–Schwarz inequality, is given in Appendix E. Thus the proposed augmented prediction is

$$\dot{z} = f_0(z, v), \tag{IV.3a}$$
$$\dot{R} = -\lambda R + q(z, v, R, s), \tag{IV.3b}$$
$$\dot{s} = -\rho s + \gamma\sqrt{R} + b_\eta. \tag{IV.3c}$$

The general notation $q(z, v, R, s)$ permits dependence on the nominal input v; the present velocity example has none explicitly. Assumption 2 states the required validity of this forcing allowance over the predicted tube.

Assumption 2 (Certificate validity on the operating domain)**.** *Let $\mathcal{D} \subseteq \mathbb{R}^n \times \mathbb{R}^{n_u}$ be a specified state–input region. The baseline observer LMIs* (III.2) *and tracking contraction condition* (IV.1) *hold on a neighborhood of $\mathcal{D}$. The disturbance-set model is valid there. For any predicted (z, v, R, s) and errors satisfying $e^\top P e \le R$ and $\|\delta\|_P \le s$, set $\hat{x} = z + \delta$, $x = z + \delta + e$, and $u = v + K\delta$. The certificate applies whenever the segments*

$$\{(\hat{x} + \theta e, u) : 0 \le \theta \le 1\},$$
$$\{(z + \theta\delta, v + \theta K\delta) : 0 \le \theta \le 1\}$$

lie in $\mathcal{D}$. The scalar forcing allowance $q(z, v, R, s)$ bounds the contribution of $E_0 w_x - LF\eta$ to the estimation-error inequality for every such true state and input, with $w_x \in \mathcal{W}(x, u)$ and $\eta \in \mathcal{N}$. Section III-C constructs this allowance for the velocity-dependent example. Under these conditions, the derivations in Sections III and IV-A give the estimation-error inequality (III.19)*, with the forcing allowance $q(z, v, R, s)$, and the tracking-error inequality* (IV.2)*.*

The two segments connect true and estimated states at a common input, and nominal and estimated states with the feedback input, respectively. To keep these certificates applicable, the MPC imposes the domain-containment condition that both segments lie in $\mathcal{D}$ for every error pair in its predicted tube. This is a constraint on the prediction, distinct from the certificate inequalities and from the physical state–input constraints.

Assumption 3 (Admissible radius box and initial bounds)**.** *The forcing bound $q(z, v, R, s)$ is locally Lipschitz in its arguments on the set of admissible nominal states, inputs, and radius pairs, and satisfies $0 < q_{\min} \le q \le q_{\max}$. Define*

$$0 < R_{\min} \le q_{\min}/\lambda, \quad R_{\max} = q_{\max}/\lambda, \quad s_{\max} = (\gamma\sqrt{R_{\max}} + b_\eta)/\rho. \tag{IV.4}$$

The pair of radius variables (R, s) is constrained to the rectangle

$$\begin{aligned}\mathcal{B} &= [R_{\min}, R_{\max}] \times [0, s_{\max}] \\ &= \{(R, s) : R_{\min} \le R \le R_{\max},\ 0 \le s \le s_{\max}\}.\end{aligned}$$

These variables bound the errors through $e^\top P e \le R$ and $\|\delta\|_P \le s$. An available initial prior $\bar{R}_0 \in [R_{\min}, R_{\max}]$ satisfies $V_{e,0} \le \bar{R}_0$. The observer state evolves continuously according to the observer dynamics in Section II, including across MPC updates.

The upper radii provide uniform error margins for the terminal construction. Local Lipschitz continuity gives a well-posed radius evolution. Boundedness follows separately from invariance of $\mathcal{B}$, while actual-error containment is established in Lemma 1. The radius dynamics point inward or tangent at all four boundaries of $\mathcal{B}$; the boundary calculations are given in Appendix F. Thus the radius pair cannot leave this rectangle along admissible nominal trajectories: if $(R_0, s_0) \in \mathcal{B}$, then $(R_t, s_t) \in \mathcal{B}$ for all subsequent times on which the assumptions hold. Since $\frac{d}{dR}\sqrt{R} = \frac{1}{2\sqrt{R}}$, the derivative is unbounded as $R \to 0^+$. Restricting $R \ge R_{\min} > 0$ bounds this derivative and ensures that the square-root term is Lipschitz on the admissible interval.

Lemma 1 (Tube containment)**.** *Under Assumptions 1–3, consider a prediction satisfying the domain-containment conditions of Section IV-B throughout an applied interval. The estimation-error inequality* (III.19) *and tracking-error inequality* (IV.2)*, together with initial bounds $V_{e,t} \le R_{0|t}$ and $\left\|\hat{x}_t - z_{0|t}\right\|_P \le s_{0|t}$, imply*

$$V_{e,t+\tau} \le R_{\tau|t}, \qquad \left\|\hat{x}_{t+\tau} - z_{\tau|t}\right\|_P \le s_{\tau|t}, \qquad 0 \le \tau \le h.$$

Proof. On $V_e = R$, the derivative of $V_e - R$ is nonpositive. Subtracting (IV.3c) from (IV.2) at $\|\delta\|_P = s$, with $V_e \le R$, gives $\frac{d}{dt}(\|\delta\|_P - s) \le \gamma(\sqrt{V_e} - \sqrt{R}) \le 0$. While the errors lie within the predicted tube, the domain-containment constraints keep the relevant state–input segments inside the certificate domain, so the observer and tracking derivative inequalities remain valid. The comparison argument under the stated regularity conditions prevents an outward crossing of either error boundary. Thus containment persists throughout the applied interval. □

For identical nominal trajectories and initial radii, the local estimation-error forcing bound $q_{\text{local,t}} \le q_0$, where $q_0 = \lambda\epsilon^2$ is the global baseline forcing bound, yields $R_{\text{local,t}} \le R_{\text{global,t}}$ and $s_{\text{local,t}} \le s_{\text{global,t}}$. Thus the local formulation produces no larger predicted error bounds under these common conditions. The derivation is given in Appendix G.

B Tightened constraints and MPC initialization

For each state–input constraint $g_j(x, u) \le 0$, impose the worst-case condition over the predicted error tube:

$$\sup_{\|e\|_P \le \sqrt{R},\ \|\delta\|_P \le s} g_j(z + \delta + e, v + K\delta) \le 0. \tag{IV.5}$$

For an affine constraint $a_j^\top x + b_j^\top u \le c_j$, substituting $x = z + \delta + e$ and $u = v + K\delta$ gives the uncertain contribution $(a_j^\top + b_j^\top K)\delta + a_j^\top e$. With $T^\top T = P$, maximization over the tracking-error ball gives

$$\max_{\|\delta\|_P \le s} (a_j^\top + b_j^\top K)\delta = s \left\|(a_j^\top + b_j^\top K)T^{-1}\right\|_2.$$

Similarly, maximization over the estimation-error ball gives

$$\max_{\|e\|_P \le \sqrt{R}} a_j^\top e = \sqrt{R} \left\|a_j^\top T^{-1}\right\|_2.$$

These maxima follow from the Cauchy–Schwarz inequality and are attained in the corresponding maximizing directions. Since the two errors range independently over their balls, the two margins add. Define the tightening constants

$$c_{s,j} := \left\|(a_j^\top + b_j^\top K)T^{-1}\right\|_2, \qquad c_{e,j} := \left\|a_j^\top T^{-1}\right\|_2.$$

For fixed affine constraints, metric P, and gain K, these constants can be computed offline. The resulting tightened constraint is

$$a_j^\top z + b_j^\top v + \underbrace{c_{s,j}s}_{\text{tracking-error margin}} + \underbrace{c_{e,j}\sqrt{R}}_{\text{estimation-error margin}} \le c_j. \tag{IV.6}$$

This expression evaluates the supremum exactly over the chosen product of estimation and tracking error balls. The margins vary along the prediction through R and s. Using the time notation in Section II, the tightened state–input constraints (IV.5) (or (IV.6) for affine constraints), certificate-domain conditions in Assumption 2, and radius bounds $(R_{\tau|t}, s_{\tau|t}) \in \mathcal{B}$ are imposed for every $\tau \in [0, H_p]$. At update t, minimize the predicted horizon cost

$$J_t = \int_0^{H_p} \ell(z_{\tau|t}, v_{\tau|t})\, d\tau + V_f(z_{H_p|t}), \tag{IV.7}$$

where ℓ is the stage-cost rate, its integral is the accumulated running cost, and V_f is the terminal cost. Initialize $R_{0|0} = \bar{R}_0$ using Assumption 3. At subsequent updates, retain the previous plan's propagated bound:

$$R_{0|t+h} = R_{h|t}. \tag{IV.8}$$

The prediction satisfies (IV.3), the constraints above, and

$$\left\|\hat{x}_t - z_{0|t}\right\|_P \le s_{0|t}, \quad (z_{H_p|t}, R_{H_p|t}, s_{H_p|t}) \in \Omega_f. \tag{IV.9}$$

The initial nominal center $z_{0|t}$ is free. For the selected feasible plan, apply

$$u_{t+\tau} = v_{\tau|t} + K(\hat{x}_{t+\tau} - z_{\tau|t}), \qquad 0 \le \tau < h,$$

and initialize the next prediction with $R_{0|t+h} = R_{h|t}$. Lemma 1 ensures $V_{e,t+h} \le R_{h|t}$, so this remains a valid estimation-error bound. Theorem 1 establishes recursive feasibility and constraint satisfaction without requiring optimality. Superscript $*$ denotes optimal plans and the minimum horizon cost in the cost-decrease inequality (IV.11) of Theorem 1.

C Terminal Ingredients and Recursive Feasibility

Assumption 4 (Terminal ingredients)**.** *There exist a nonempty nominal terminal set Z_f and controller κ_f such that $\dot{z} = f_0(z, \kappa_f(z))$ leaves Z_f invariant. For every $z \in Z_f$, with nominal input $v = \kappa_f(z)$, the tightened state–input constraints* (IV.5) *and the certificate-domain conditions in Assumption 2 hold at $R = R_{\max}$ and $s = s_{\max}$. For the cost assertion additionally assume $\ell, V_f \ge 0$ and*

$$\nabla V_f(z) f_0(z, \kappa_f(z)) \le -\ell(z, \kappa_f(z)). \tag{IV.10}$$

Then $\Omega_f = Z_f \times \mathcal{B}$ is invariant under terminal control and the radius dynamics. The terminal set is certified using the largest admissible estimation and tracking radii. Since the smaller error balls are contained in these largest ones, this ensures terminal constraint satisfaction and certificate-domain containment for every $(R, s) \in \mathcal{B}$, while the predictions along the horizon retain their local, time-varying radii.

Theorem 1 (Recursive Feasibility and Robust Constraint Satisfaction)**.** *Suppose Assumptions 1–4 hold and the initial MPC problem is feasible. Applying a feasible plan satisfying* (IV.9) *and initializing* $R_{0|0} = \bar{R}_0$ *and* $R_{0|t+h} = R_{h|t}$ *at subsequent updates gives recursive feasibility and continuous-time robust constraint satisfaction for all disturbances in the assumed sets. If* (IV.10) *holds and the selected feasible plans attain the minimum horizon cost among all feasible plans, then*

$$J^*_{t+h} \le J^*_t - \int_0^h \ell(z^*_{\tau|t}, v^*_{\tau|t})\, d\tau. \tag{IV.11}$$

Proof.

Part 1: Recursive feasibility.

Suppose the plan $(z_{\tau|t}, v_{\tau|t}, R_{\tau|t}, s_{\tau|t})$ is feasible at update t. For $0 \le \tau \le H_p - h$, define

$$z^c_{\tau|t+h} = z_{\tau+h|t}, \qquad v^c_{\tau|t+h} = v_{\tau+h|t},$$
$$R^c_{\tau|t+h} = R_{\tau+h|t}, \qquad s^c_{\tau|t+h} = s_{\tau+h|t}.$$

Evaluating the shifted-candidate definition at $\tau = 0$ gives

$$z^c_{0|t+h} = z_{h|t}, \qquad R^c_{0|t+h} = R_{h|t}, \qquad s^c_{0|t+h} = s_{h|t}.$$

The estimation-radius update rule (IV.8) sets $R_{0|t+h} = R_{h|t}$, so the candidate has the required initial radius $R^c_{0|t+h} = R_{0|t+h}$. Lemma 1, applied over the previous control interval, ensures $V_{e,t+h} \le R_{h|t}$ and $\left\|\hat{x}_{t+h} - z_{h|t}\right\|_P \le s_{h|t}$. Using the shifted initial center and tracking radius therefore yields

$$\left\|\hat{x}_{t+h} - z^c_{0|t+h}\right\|_P = \left\|\hat{x}_{t+h} - z_{h|t}\right\|_P \le s_{h|t} = s^c_{0|t+h}.$$

Thus the candidate satisfies the prescribed estimation-radius initialization and the initial tracking bound in (IV.9). On this shifted interval the candidate reproduces the previous feasible trajectory. The coupled nominal and tube dynamics (IV.3) and prediction constraints have no explicit time dependence. Consequently, the shifted candidate continues to satisfy those dynamics, the tightened constraints (IV.5), certificate-domain conditions, and radius bounds.

At $\tau = H_p - h$, the candidate reaches the previous terminal point in Ω_f. Over the remaining h units of time, apply $v^c_{\tau|t+h} = \kappa_f(z^c_{\tau|t+h})$ and continue the nominal and radius dynamics. Terminal invariance under Assumption 4 and invariance of $\mathcal{B}$ keep this continuation in Ω_f, where all tightened and domain constraints hold. Its endpoint therefore satisfies the new terminal condition. This constructs a feasible plan at update $t + h$; induction from initial feasibility proves recursive feasibility.

Part 2: Robust constraint satisfaction.

During each applied interval, Lemma 1 gives $\|e\|_P \le \sqrt{R_{\tau|t}}$ and $\|\delta\|_P \le s_{\tau|t}$. The actual pair satisfies $x_{t+\tau} = z_{\tau|t} + \delta + e$ and $u_{t+\tau} = v_{\tau|t} + K\delta$, and hence belongs to the set over which (IV.5) takes its supremum. Consequently,

$$g_j(x_{t+\tau}, u_{t+\tau}) \le \sup_{\|e\|_P \le \sqrt{R_{\tau|t}},\ \|\delta\|_P \le s_{\tau|t}} g_j(z_{\tau|t} + \delta + e, v_{\tau|t} + K\delta) \le 0.$$

This holds throughout every applied interval and for all admissible disturbances.

Part 3: Optimal-cost decrease.

Now take an optimal plan at update t. Let $z_{f,\sigma}$, $0 \le \sigma \le h$, be the appended terminal trajectory, with $z_{f,0} = z^*_{H_p|t}$ and input $\kappa_f(z_{f,\sigma})$. The shifted candidate has cost

$$J^c_{t+h} = \int_h^{H_p} \ell(z^*_{\tau|t}, v^*_{\tau|t})\, d\tau + \int_0^h \ell(z_{f,\sigma}, \kappa_f(z_{f,\sigma}))\, d\sigma + V_f(z_{f,h}).$$

Integrating (IV.10) along the terminal continuation gives

$$\int_0^h \ell(z_{f,\sigma}, \kappa_f(z_{f,\sigma}))\, d\sigma + V_f(z_{f,h}) \le V_f(z^*_{H_p|t}).$$

The terminal-cost decrease therefore covers the accumulated stage cost of the appended interval. Substituting this bound and comparing with (IV.7) yields

$$\begin{aligned} J^c_{t+h} &\le \int_h^{H_p} \ell(z^*_{\tau|t}, v^*_{\tau|t})\, d\tau + V_f(z^*_{H_p|t}) \\ &= J^*_t - \int_0^h \ell(z^*_{\tau|t}, v^*_{\tau|t})\, d\tau. \end{aligned}$$

Since the next optimal cost is no greater than the cost of this feasible candidate, $J^*_{t+h} \le J^c_{t+h}$, which establishes (IV.11). □

The shifted-and-extended candidate provides a feasible fallback. If a suboptimal solver returns a feasible plan whose horizon cost does not exceed the candidate cost, the same argument establishes the cost-decrease inequality for the selected plans.

Remark 1 (Relation to [1]). Our formulation follows the free-center approach in Section IV-B of [1]: the optimizer selects the initial nominal center subject to the tracking-error bound. The shifted center $z^c_{0|t+h} = z_{h|t}$ belongs to the feasible candidate constructed in the proof of Theorem 1; it is not imposed on the newly optimized plan. The estimation radius is initialized using (IV.8). Section IV-C of [1] instead fixes the initial nominal center to the current estimate and establishes recursive feasibility under its corresponding terminal conditions. Here, we retain the free-center architecture and incorporate the state-dependent forcing derived in Section III.

The horizon-cost decrease in (IV.11) bounds the accumulated nominal running cost. By (II.3), the true-state deviation satisfies

$$\|x - x_{\mathrm{ref}}\|_P \le \|z - x_{\mathrm{ref}}\|_P + \sqrt{R} + s.$$

Consequently, if the applied nominal trajectory converges to x_{ref}, then

$$\limsup_{t\to\infty} \|x_t - x_{\mathrm{ref}}\|_P \le \sqrt{R_{\max}} + s_{\max},$$

where t denotes continuous time. Nominal convergence is an additional condition here; it is not established by the cost-decrease inequality alone.

V Numerical Example: Falcon Model

A Model and reproduced offline design

The example uses the repository [3] and the saved Falcon design from 10 September 2026. The 12-state model consists of position, attitude, translational velocity, and body angular velocity, with four rotor inputs. The nominal translational drag matrix is zero in this instance. Its Jacobians are therefore independent of translational velocity; this does not restrict the controller to hovering. The velocity domain is $[-2, 2]^3$ m/s, giving $v_{\max} = \sqrt{12}$ m/s.

The original 1024-point Jacobian grid varies four rotor inputs, three angles, and three body rates at two endpoints each. Positions and translational velocities are zero at these points. This grid alone does not certify the continuous domain of the remaining nonlinear coordinates.

The reproduced robust-design and terminal-cost SDPs returned `OPTIMAL`, with objectives 1494.858118 and 38.564610, respectively. The first is a weighted combination of normalized

squared tightening coefficients and squared estimation radius, not an optimized contraction rate. The second minimizes the trace of the terminal-cost matrix. Solver success does not ensure useful tightened constraints: the reproduced motor-input half-width was approximately 0.1200, whereas the associated global margin was 0.4619, leaving negative available width. Consequently, this design has not supplied the nonempty terminal set required by Theorem 1.

B Prescribed local uncertainty and parameters

For the Falcon state ordering (position, attitude, translational velocity, body angular velocity), the velocity selector and disturbance scaling matrix are

$$S_v = \begin{bmatrix} 0_{3\times 6} & I_3 & 0_{3\times 3} \end{bmatrix},$$
$$D(a) = \operatorname{diag}(I_6, aI_3, I_3).$$

The scaled disturbance components enter the translational acceleration equations. Only the centered acceleration disturbance components 7–9 are scaled. Other process-disturbance components, measurement noise, and the nominal bias remain unchanged. We use $\beta = 0.25$ and the velocity law in Section III-C. This is a synthetic uncertainty model, not a new identification or UQ result. The experiment reuses the saved $X, P, L, K, W, H, \lambda$ and solves no new SDP. The factor 0.99

Table 1: Saved certificate and local-envelope parameters (rounded).

Quantity	Value
Observer decay coefficient λ	0.497029
Original uncalibrated ϵ	0.07442638619
Global forcing q_0	0.002753187883
$g(1)$	0.00148773
$g(\beta)$	0.00104178
$\Delta = 0.99[g(1) - g(\beta)]$	0.000441487
$q(\beta)$	0.0023117
Condition number of Q	3474.67

retains a numerical cushion but does not repair an infeasible baseline certificate or provide a rigorous floating-point error bound.

C Radius propagation experiment

A prescribed nominal speed is 0 m/s on $[0, 8)$ s, 2.5 m/s on $[8, 16)$ s, and 0 m/s on $[16, 32]$ s. Initialize $R_0 = \epsilon^2$, $s_0 = 0$. Both cases use identical metrics, gains, initial bounds, tracking dynamics, and noise; only the local forcing versus q_0 changes. The global estimation radius equals ϵ by (III.5). The final estimation-radius reduction is $100(1 - r_{\text{local}}/\epsilon) = 6.68019\%$. This is

Table 2: Final-time comparison at 32 s; these are bounds, not measured errors.

Quantity	Global	State-dependent
Estimation radius r	0.07442638619	0.06945456129
Tracking radius s	0.392384	0.367532

neither a time-average improvement nor a universal performance guarantee. The speed schedule is not a dynamically feasible quadrotor trajectory, and no MPC optimization or closed-loop plant simulation is performed. The earlier separate-norm prototype produced a global equilibrium radius of approximately 0.105718 and a final local radius of 0.0979251, both above the original

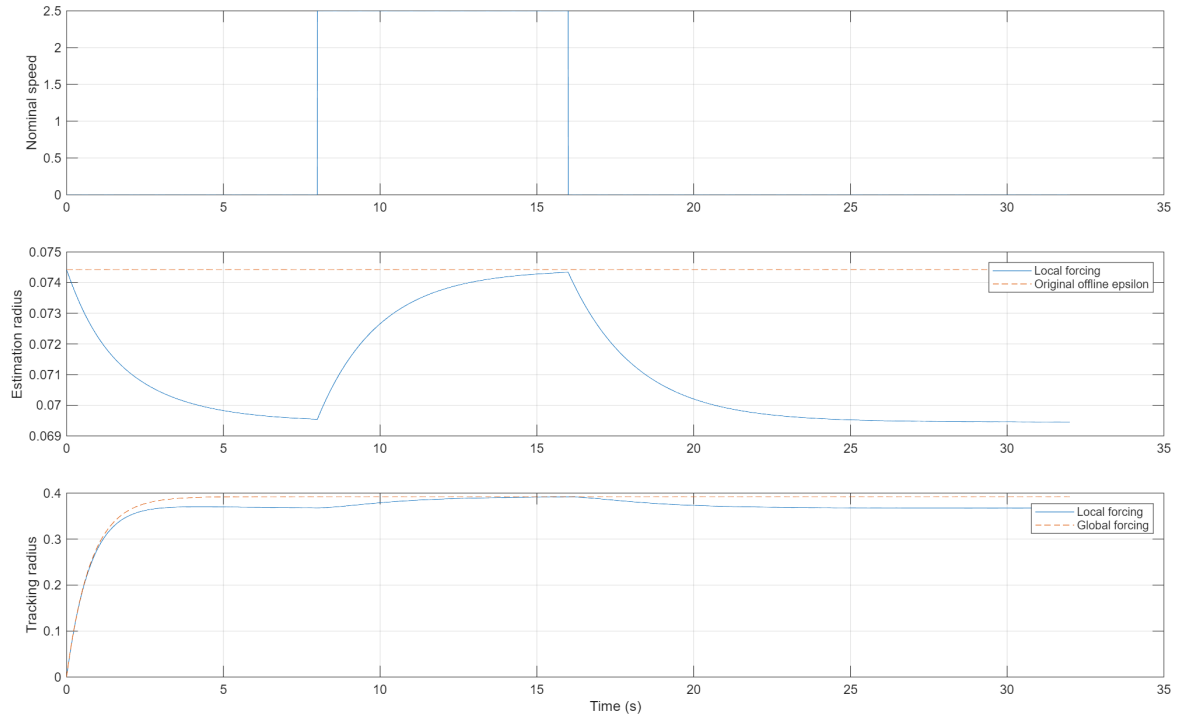


Figure 1: Recorded tube-ODE experiment with a prescribed low/high/low nominal-speed schedule. The figure is copied from the saved run, not generated by a closed-loop simulator.

0.0744264. These values belong to a different bound construction and do not indicate empirical calibration of the original ϵ. The quadratic-envelope approach restores a baseline-equivalent comparison.

D Numerical checks and limitations

A numerical MPC implementation that checks constraints only at selected prediction times must additionally establish constraint satisfaction between those times. The continuous-time formulation in Section IV-B imposes them throughout the horizon. The current experiment checks all 4096 original process vertices, all noise vertices, the 1024 original dynamics points, and all process vertices at 31 scale values. It additionally evaluates 1968 admissible random nonlinear error configurations. Table 3 reports the saved residuals. The saved feasibility tolerance is 10^{-6}.

Table 3: Numerical audit; positive matrix residuals are tolerated numerically, not exact certificates.

Check	Maximum residual
Original process-envelope LMI	$+5.50832\,10^{-8}$
Original noise-envelope LMI	$+5.57035\,10^{-8}$
Observer dynamics LMI	$+5.45468\,10^{-8}$
Tracking LMI	-0.0030671
Local Schur envelope	$+5.63813\,10^{-8}$
Sampled nonlinear derivative excess	-0.00229595

Small positive residuals mean feasibility within that numerical tolerance, not exact semidefinite feasibility. The convex-chord argument covers intermediate scales analytically only conditional on exact baseline inequalities. Finite Jacobian and nonlinear samples do not prove the domain-wide inequalities required in Assumption 2. No closed-loop feasibility, safety, or stability claim for Falcon follows from these tests.

VI Conclusion

A fixed observer certificate can support locally reduced process forcing through a Schur-complement envelope construction. Coupling this forcing to a tracking radius yields a candidate state-dependent output-feedback MPC formulation with conditional recursive-feasibility and safety guarantees. The Falcon radius experiment demonstrates a smaller local bound relative to a baseline-equivalent global propagation. Remaining work is to certify the continuous domain and numerical margins, validate the local uncertainty model, construct compatible nonempty terminal ingredients, and then implement and evaluate the closed-loop controller.

References

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[2] D. Benders, J. Köhler, R. Babuška, J. Alonso-Mora, and L. Ferranti, "From Data to Safe Mobile Robot Navigation: An Efficient and Modular Robust MPC Design Pipeline," arXiv:2508.07045v1, 2025. `https://arxiv.org/abs/2508.07045`.

[3] D. Benders et al., ROHMPC code repository, `https://github.com/dbenders1/rohmpc`. Numerical inputs and saved outputs from the local reproduction of 10 September 2026.

APPENDIX

A From Observer LMIs to the Error Inequality

Use the error dynamics $\dot{e} = A_o e + b$, with $b = E_0 w - LF\eta$ and the segment-averaged A_o defined in Section III-A. The following calculation expands the implication from (III.2) to (III.3). Rewrite the first two inequalities in (III.2) as $\mathcal{S}(E_0 w) - W \preceq 0$ and $\mathcal{S}(-LF\eta) - H \preceq 0$, and add the third. The auxiliary matrices cancel:

$$\underbrace{\mathcal{S}(E_0 w) - W}_{\preceq 0} + \underbrace{\mathcal{S}(-LF\eta) - H}_{\preceq 0} + \underbrace{W + H + \operatorname{diag}(XA_o^\top + A_o X + \lambda X, -q_0)}_{\preceq 0} \preceq 0.$$

Since $\mathcal{S}$ is linear,

$$\mathcal{S}(E_0 w) + \mathcal{S}(-LF\eta) = \mathcal{S}(b),$$

so the resulting matrix inequality is

$$M := \begin{bmatrix} XA_o^\top + A_o X + \lambda X & b \\ b^\top & -q_0 \end{bmatrix} \preceq 0.$$

Negative semidefiniteness means $\xi^\top M \xi \leq 0$ for every vector ξ. Choosing $\xi = [Pe; 1]$ and using $PX = XP = I$ and constant P gives

$$\begin{aligned} \begin{bmatrix} Pe \\ 1 \end{bmatrix}^\top M \begin{bmatrix} Pe \\ 1 \end{bmatrix} &= e^\top P(XA_o^\top + A_o X + \lambda X)Pe + 2e^\top Pb - q_0 \\ &= e^\top (A_o^\top P + PA_o)e + 2e^\top Pb + \lambda e^\top Pe - q_0 \\ &= \dot{V}_e + \lambda V_e - q_0 \leq 0. \end{aligned}$$

This establishes the error inequality (III.3). Equivalently, define the invertible matrix $G = \operatorname{diag}(P, 1)$. The congruence transformation of M by G is

$$G^\top M G = \begin{bmatrix} A_o^\top P + PA_o + \lambda P & Pb \\ b^\top P & -q_0 \end{bmatrix} \preceq 0. \tag{A.1}$$

Indeed, $v^\top G^\top M G v = (Gv)^\top M (Gv) \leq 0$ for every v; invertibility makes the two matrix inequalities equivalent. Evaluating the transformed matrix at $[e; 1]$ is identical to evaluating M at $[Pe; 1]$, because $G[e; 1] = [Pe; 1]$. These inequalities hold for every admissible w, η.

B Vertex Maximum, Convexity, and Chord Bound

Recall $\phi(w, a)$, its vertex values $\phi_i(a)$, and $g(a)$ from Section III-B.

Maximum over the disturbance box. For fixed a, the Hessian with respect to w is

$$\nabla_w^2 \phi(w, a) = 2D(a)^\top E_0^\top Q^{-1} E_0 D(a) \succeq 0,$$

so $\phi(\cdot, a)$ is convex. Any $w \in \mathcal{W}_0$ can be written as $w = \sum_i \theta_i w^{(i)}$ with $\theta_i \geq 0$ and $\sum_i \theta_i = 1$. Therefore,

$$\phi(w, a) \leq \sum_i \theta_i \phi(w^{(i)}, a) \leq \max_i \phi_i(a) = g(a).$$

Since every box point is a convex combination of its vertices and the vertices belong to the box, $g(a) = \max_{w \in \mathcal{W}_0} \phi(w, a)$. Thus the vertex maximum bounds the entire disturbance box, including its interior.

Convexity in the scheduling amplitude. For a fixed vertex, the affine dependence of $D(a)$ permits the representation

$$E_0 D(a) w^{(i)} - h_w = a p_i + r_i, \qquad \phi_i''(a) = 2 p_i^\top Q^{-1} p_i \geq 0,$$

where p_i, r_i are fixed vectors. Each vertex function is consequently convex in a, and their pointwise maximum g is convex as well, although it need not be differentiable where the maximizing vertex changes. Nestedness of $E_0 D(a) \mathcal{W}_0$ also gives $g(\beta) \leq g(1)$.

Endpoint chord and secant slopes. A convex function lies below the chord joining its endpoint values. Writing $a = \frac{1-a}{1-\beta}\beta + \frac{a-\beta}{1-\beta} 1$ for $a \in [\beta, 1]$ yields

$$g(a) \leq \frac{1-a}{1-\beta} g(\beta) + \frac{a-\beta}{1-\beta} g(1).$$

This is the chord bound (III.11). For $\beta < a < 1$, the same inequality can be expressed as a comparison of secant slopes:

$$\frac{g(1) - g(a)}{1-a} \geq \frac{g(1) - g(\beta)}{1-\beta}.$$

The secant ending at 1 and starting at a has slope at least as large as the secant from β to 1. Hence the chord provides a guaranteed reduction from the global value $g(1)$, without requiring derivatives of g.

C Velocity Bound over the Error Tube

Recall $c_v = \left\| S_v T^{-1} \right\|_2$, $T^\top T = P$, and $x = z + \delta + e$ from Section III-C. Using the Cauchy–Schwarz inequality and the induced matrix norm bound, we obtain

$$\begin{aligned} \|S_v e\|_2 &= \left\| S_v T^{-1} (Te) \right\|_2 \\ &\leq \left\| S_v T^{-1} \right\|_2 \|Te\|_2 = c_v \|e\|_P \leq c_v \sqrt{R}. \end{aligned}$$

The same argument gives $\|S_v \delta\|_2 \leq c_v \|\delta\|_P \leq c_v s$. Starting from the state decomposition $x = z + \delta + e$, the triangle inequality gives

$$\begin{aligned} \|S_v x\|_2 &= \|S_v z + S_v \delta + S_v e\|_2 \\ &\leq \|S_v z\|_2 + \|S_v \delta\|_2 + \|S_v e\|_2 \\ &\leq \|S_v z\|_2 + c_v (s + \sqrt{R}). \end{aligned}$$

D Comparison with Separate-Norm Disturbance Bounds

Retaining the model, metric, and gains from Section II and the uncertainty sets from Section III-B, consider an alternative estimation-error bound obtained by bounding the process-disturbance and measurement-noise contributions separately. Derivative inequalities for error norms are understood to hold almost everywhere in time. If the disturbance-free observer error satisfies $\frac{d}{dt} \|e\|_P \leq -\rho_o \|e\|_P$ for a contraction rate $\rho_o > 0$, a radius bound can be propagated as

$$\dot{r} = -\rho_o r + b_{\text{rest}} + a_{\text{up}} b_{\text{acc}} + b_\eta. \tag{D.1}$$

Here $b_{\text{acc}} = \max \|T E_{\text{acc}} w_{\text{acc}}\|_2$ bounds the unscaled acceleration-disturbance contribution, b_{rest} bounds the remaining process contribution, and $b_\eta = \max_{\eta \in \mathcal{N}} \|TLF\eta\|_2$ bounds measurement noise. Each maximum is taken over its corresponding admissible set.

For vector contributions $d_j(\omega_j)$ with $\omega_j \in \Omega_j$, this construction uses

$$\max_{\omega \in \prod_j \Omega_j} \left\| \sum_j d_j(\omega_j) \right\|_P \leq \sum_j \max_{\omega_j \in \Omega_j} \|d_j(\omega_j)\|_P .$$

The triangle inequality and separate maximizations discard directional information: the individual largest contributions need not align to attain the sum of their norms. Thus this approach can be more conservative and need not recover the original SDP radius at $a = 1$. Proposition 1 retains the original quadratic certificate and recovers its global baseline at $a = 1$.

E Derivation of the Tracking-Error Bound

Recall $\delta = \hat{x} - z$, $e = x - \hat{x}$, and $u = v + K\delta$. Subtracting the nominal dynamics from the observer dynamics gives

$$\dot{\delta} = f_0(\hat{x}, u) - f_0(z, v) + LCe + LF\eta. \tag{E.1}$$

For fixed z, v, δ, parameterize the straight-line state–input segment by

$$x_\theta = z + \theta\delta, \qquad u_\theta = v + \theta K\delta, \qquad 0 \leq \theta \leq 1.$$

Here θ parameterizes a segment in state–input space, not time. Its endpoints are (z, v) and $(\hat{x}, u)$. With $A = \partial f_0/\partial x$ and $B = \partial f_0/\partial u$, integration along this segment yields

$$f_0(\hat{x}, u) - f_0(z, v) = \int_0^1 \big[A(x_\theta, u_\theta) + B(x_\theta, u_\theta)K\big]\delta \, d\theta \tag{E.2}$$

$$= \bar{A}_K \delta, \qquad \bar{A}_K := \int_0^1 \big[A(x_\theta, u_\theta) + B(x_\theta, u_\theta)K\big] d\theta. \tag{E.3}$$

The domain assumptions ensure the contraction inequality holds at every point of this segment. Integrating it and using constant P gives

$$\bar{A}_K^\top P + P\bar{A}_K \preceq -2\rho P. \tag{E.4}$$

Define $d_o := LCe + LF\eta$. Then (E.1) becomes $\dot{\delta} = \bar{A}_K\delta + d_o$. For $\ell_\delta := \|\delta\|_P = \sqrt{\delta^\top P\delta} > 0$,

$$\dot{\ell}_\delta = \frac{\delta^\top P\dot{\delta}}{\ell_\delta} \tag{E.5}$$

$$= \frac{\delta^\top P\bar{A}_K\delta}{\ell_\delta} + \frac{\delta^\top P d_o}{\ell_\delta}. \tag{E.6}$$

The contraction contribution satisfies

$$\delta^\top P\bar{A}_K\delta = \frac{1}{2}\delta^\top(P\bar{A}_K + \bar{A}_K^\top P)\delta \leq -\rho\ell_\delta^2.$$

For the forcing contribution, $T^\top T = P$ and the Cauchy–Schwarz inequality imply

$$\frac{\delta^\top P d_o}{\ell_\delta} = \frac{(T\delta)^\top (T d_o)}{\|T\delta\|_2} \leq \|T d_o\|_2 .$$

The triangle inequality and the definitions of γ and b_η give

$$\|T d_o\|_2 \leq \|TLCe\|_2 + \|TLF\eta\|_2 \tag{E.7}$$

$$\leq \left\|TLCT^{-1}\right\|_2 \|Te\|_2 + \max_{\eta \in \mathcal{N}} \|TLF\eta\|_2 \tag{E.8}$$

$$= \gamma\sqrt{V_e} + b_\eta. \tag{E.9}$$

Combining these estimates establishes

$$\frac{d}{dt}\|\delta\|_P \le -\rho\|\delta\|_P + \gamma\sqrt{V_e} + b_\eta, \tag{E.10}$$

which is (IV.2). The calculation involving division by ℓ_δ applies when $\delta \neq 0$. Along absolutely continuous trajectories, the norm is absolutely continuous and its derivative is zero almost everywhere on the set where the norm vanishes. Since the forcing bound is nonnegative, the same inequality therefore holds almost everywhere in time, including that zero set.

Finally, $V_e \le R$ gives the comparison dynamics

$$\dot{s} = -\rho s + \gamma\sqrt{R} + b_\eta, \qquad s_0 \ge \|\delta_0\|_P .$$

Subtracting this equation from the norm inequality and applying an integrating factor yields

$$\|\delta_t\|_P - s_t \le e^{-\rho t}\big(\|\delta_0\|_P - s_0\big) \le 0,$$

provided the estimation bound and certificate-domain conditions remain valid, as required in the coupled containment argument.

F Invariance of the Radius Box

Recall the rectangle $\mathcal{B}$ and the bounds $q_{\min} \le q \le q_{\max}$ in Assumption 3. The coupled radius equations (IV.3) satisfy the following conditions on the four boundaries of $\mathcal{B}$:

$$\begin{aligned}
R = R_{\min}: &\quad \dot{R} \ge -\lambda R_{\min} + q_{\min} \ge 0,\\
R = R_{\max}: &\quad \dot{R} \le -\lambda R_{\max} + q_{\max} = 0,\\
s = 0: &\quad \dot{s} = \gamma\sqrt{R} + b_\eta \ge 0,\\
s = s_{\max}: &\quad \dot{s} \le -\rho s_{\max} + \gamma\sqrt{R_{\max}} + b_\eta = 0.
\end{aligned}$$

Each boundary derivative is inward-pointing or tangent. Under the stated regularity conditions, the box is therefore invariant along admissible nominal trajectories: a radius pair initialized in $\mathcal{B}$ remains there while the assumptions hold.

G Comparison of Local and Global Radius Bounds

Consider the same nominal state and input trajectories z_t, v_t, identical initial radius values, and the same metric, gains, and constants $\lambda, \rho, \gamma, b_\eta$ in both propagations. Define the local estimation-error forcing along its radius trajectory by

$$q_{\text{local,t}} := q\big(z_t, v_t, R_{\text{local,t}}, s_{\text{local,t}}\big).$$

For the envelope construction in Proposition 1, $q_{\text{local,t}} \le q_0$, where $q_0 = \lambda\epsilon^2$ is the global baseline forcing bound. The comparison below holds on any interval on which both propagations are well-defined and their domain assumptions hold.

The squared estimation radii satisfy

$$\dot{R}_{\text{local}} = -\lambda R_{\text{local}} + q_{\text{local,t}}, \tag{G.1}$$

$$\dot{R}_{\text{global}} = -\lambda R_{\text{global}} + q_0, \qquad R_{\text{local},0} = R_{\text{global},0}. \tag{G.2}$$

Let $\Delta_R = R_{\text{local}} - R_{\text{global}}$. Subtracting the equations and multiplying by $e^{\lambda t}$ gives

$$\frac{d}{dt}\big(e^{\lambda t}\Delta_{R,t}\big) = e^{\lambda t}\big(q_{\text{local,t}} - q_0\big).$$

Integration, using $\Delta_{R,0} = 0$, yields

$$R_{\text{local,t}} - R_{\text{global,t}} = \int_0^t e^{-\lambda(t-\tau)} (q_{\text{local},\tau} - q_0)\, d\tau \leq 0. \tag{G.3}$$

The inequality follows because the exponential is positive and the forcing difference is nonpositive. This argument permits q_{local} to depend on the local radius trajectory; it only uses its upper bound by q_0.

The corresponding tracking radii satisfy

$$\dot{s}_{\text{local}} = -\rho s_{\text{local}} + \gamma\sqrt{R_{\text{local}}} + b_\eta, \tag{G.4}$$

$$\dot{s}_{\text{global}} = -\rho s_{\text{global}} + \gamma\sqrt{R_{\text{global}}} + b_\eta, \qquad s_{\text{local},0} = s_{\text{global},0}. \tag{G.5}$$

Subtracting and applying the integrating factor $e^{\rho t}$ similarly gives

$$\begin{aligned} s_{\text{local,t}} - s_{\text{global,t}} &= \gamma \int_0^t e^{-\rho(t-\tau)} \left(\sqrt{R_{\text{local},\tau}} - \sqrt{R_{\text{global},\tau}}\right) d\tau \\ &\leq 0. \end{aligned} \tag{G.6}$$

Here (G.3), nonnegative radii, and $\gamma \geq 0$ establish the final inequality. Consequently, under the common nominal trajectories and initialization, both predicted error bounds in the local formulation are no larger than their global baseline counterparts.